

Supplementary Materials for the Manuscript “An Adaptive Neuro-Fuzzy System with Integrated Feature Selection and Rule Extraction for High-Dimensional Classification Problems”

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A. The Comparison of Gate Functions

In Section III-A, three existing gate functions are mentioned, i.e., $M(\lambda) = e^{-\lambda^2}$, $M(\lambda) = 1 - e^{-\lambda^2}$ and $M(\lambda) = \frac{1}{1+e^{-\lambda}}$. In Section IV-A, we have compared the effectiveness of the proposed gate function and $M(\lambda) = e^{-\lambda^2}$ on the Wine dataset. Here we provide additional comparison considering all three gate functions also using the Wine dataset in Fig. S-1. Fig. S-1 has eight panels. From these eight figures, we can see that the proposed gate function can learn to distinguish between useful and poor features faster than the existing ones using the same learning rate. Even with lower learning rate, the proposed gate function can better distinguish between features.

B. Effect of Different Choices of the Number of Fuzzy Sets on FSRE-AdaTSK

Earlier, in Section IV-C, we have used $S = 10$ and $S = 5$ fuzzy sets for each feature in the feature selection and rule extraction phases, respectively. To demonstrate that the choice of these parameters is not very critical, here we do some experiments with number of linguistic values on each feature as 9 and 11 for the feature selection phase and 4 and 6 for the rule extraction phase. The results are reported in Table S-I. In Table S-I, the column heading (p, q) indicates the use of p linguistic values for feature selection and q linguistic values for rule extraction. Inspection of the results in the Table IV-C reveals that the performance of our proposed method is not very sensitive to the choice of the number of linguistic values in the feature selection and rule extraction phases.

TABLE S-I

EFFECT OF THE CHOICE OF THE NUMBER OF LINGUISTIC VALUES IN THE FEATURE SELECTION AND RULE EXTRACTION PHASES ON THE PERFORMANCE OF ADATSK FOR BOTH LOW AND HIGH-DIMENSIONAL DATASETS

Datasets	(9, 5)	(11, 5)	(10, 4)	(10, 6)	(10, 5)
Iris (4)	95.7 (2.5, 5.0)	95.7 (2.4, 5.0)	96.4 (2.0, 3.2)	95.9 (2.0, 5.5)	96.5 (2.1, 6.1)
Appendicitis (7)	86.4 (3.6, 3.5)	85.1 (3.8, 3.2)	85.3 (3.8, 3.5)	84.9 (3.7, 3.3)	86.3 (3.7, 3.1)
Pima (8)	75.4 (3.3, 3.2)	75.7 (3.1, 3.0)	76.2 (2.9, 3.9)	76.2 (3.0, 3.7)	75.7 (3.0, 3.2)
Yeast (8)	59.0 (4.9, 10.0)	58.3 (4.7, 10.0)	57.6 (4.6, 10.0)	57.1 (4.6, 10.0)	58.1 (4.5, 10.0)
Glass (9)	65.4 (4.1, 6.0)	65.0 (4.1, 6.0)	62.3 (4.1, 6.0)	64.7 (4.1, 6.0)	65.5 (4.1, 6.0)
Page-blocks (10)	92.2 (4.3, 10.0)	92.1 (3.9, 8.8)	92.8 (4.1, 5.6)	92.1 (4.1, 9.4)	93.4 (2.2, 5.6)
Wine (13)	96.7 (6.1, 6.2)	96.9 (6.1, 6.5)	96.5 (6.3, 8.0)	97.9 (6.4, 8.8)	97.3 (6.3, 6.3)
Heart (13)	81.9 (7.3, 2.7)	82.3 (7.1, 3.0)	82.0 (7.4, 3.8)	81.6 (7.5, 3.3)	82.5 (7.4, 2.9)
Wdbc (30)	95.3 (7.2, 4.3)	95.6 (6.9, 4.2)	95.4 (6.2, 4.7)	95.8 (6.1, 6.6)	95.4 (6.2, 4.9)
Texture (40)	97.0 (24.0, 11.0)	97.3 (22.7, 11.0)	97.5 (23.7, 11.0)	97.7 (23.8, 11.0)	97.2 (23.7, 11.0)
Spectfheart (44)	77.6 (4.4, 5.6)	78.0 (2.4, 3.9)	78.2 (3.4, 4.6)	78.1 (3.7, 4.1)	79.2 (3.4, 5.1)
Sonar (60)	75.3 (9.9, 2.0)	74.1 (9.5, 2.1)	72.6 (10.8, 2.0)	73.0 (10.9, 2.1)	73.4 (10.5, 2.1)
ORL (1024)	82.2 (53.8, 40.0)	80.0 (51.4, 40.0)	82.4 (64.3, 40.0)	84.5 (65.4, 40.0)	83.5 (63.7, 40.0)
Colon (2000)	81.0 (8.0, 5.2)	82.7 (7.0, 4.1)	81.1 (8.9, 2.1)	83.7 (9.1, 13.3)	81.6 (9.0, 6.2)
SRBCT (2038)	95.1 (11.3, 4.1)	95.9 (11.9, 4.1)	96.6 (12.5, 6.1)	95.6 (12.5, 4.1)	96.7 (12.3, 4.2)
ARP (2400)	77.8 (41.2, 33.3)	77.5 (44.9, 43.6)	71.1 (43.6, 17.4)	74.2 (45.0, 23.2)	77.7 (45.0, 76.4)
PIE (2420)	96.2 (37.4, 10.6)	95.4 (25.7, 10.3)	98.1 (38.5, 12.9)	96.4 (38.0, 10.0)	96.2 (37.9, 10.1)
Leukemia (7129)	92.4 (5.2, 2.5)	91.1 (5.1, 2.4)	93.0 (11.3, 2.3)	94.3 (11.4, 4.9)	93.4 (11.5, 4.2)
CNS (7129)	68.0 (25.9, 5.0)	69.7 (28.9, 5.5)	72.6 (52.1, 5.0)	72.6 (49.4, 5.1)	71.0 (51.9, 15.1)

C. The Training Results and Convergence Plots of FSRE-AdaTSK

Tables S-II and S-III report both the training and test accuracies on the low-dimensional and high-dimensional datasets, respectively. From Table S-II, we can observe that the training accuracies are similar to the test accuracies on each dataset, which suggests that the model is well-trained. While from Table S-III, we can see that for a few datasets, particularly for ORL, ARP, and CNS, the difference between the training and test accuracies is quite large. For each of these datasets, the average

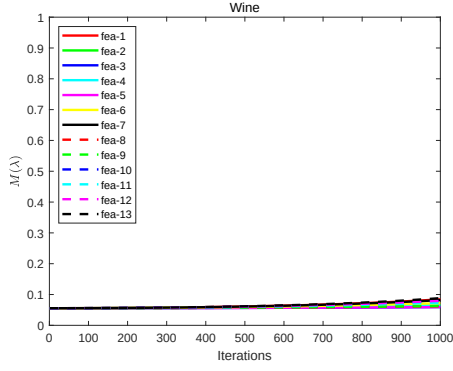
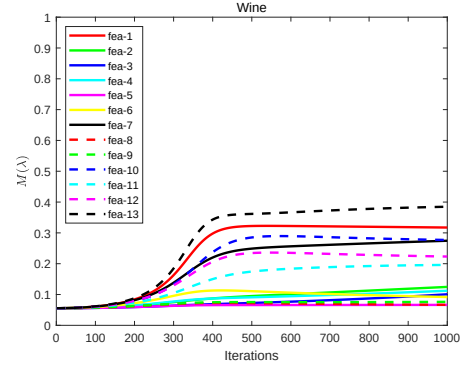
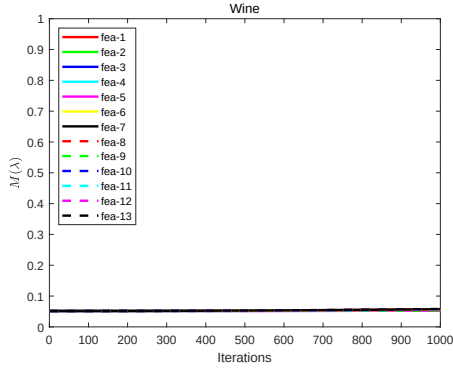
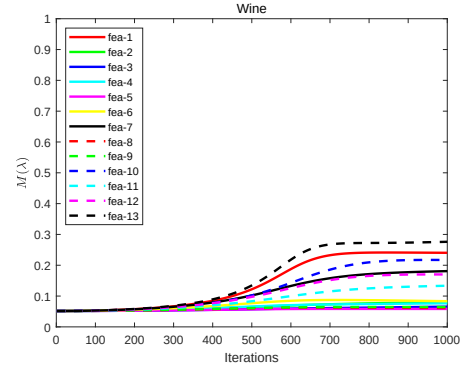
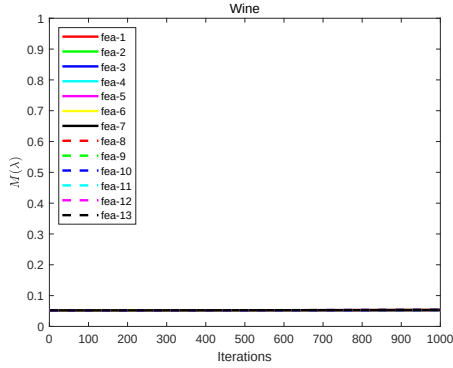
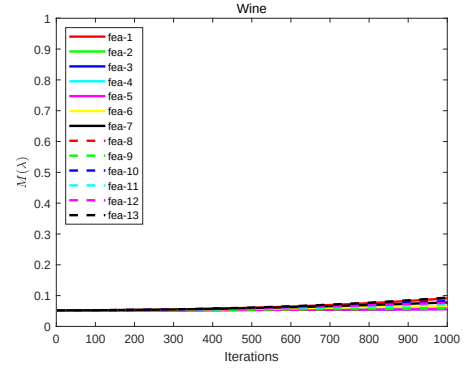
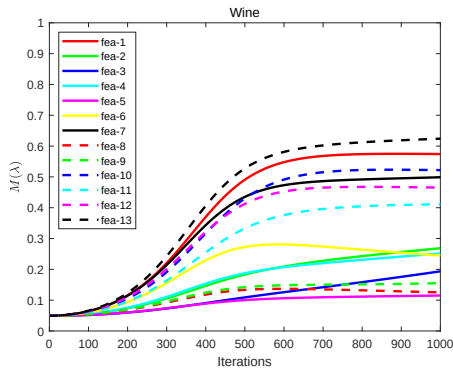
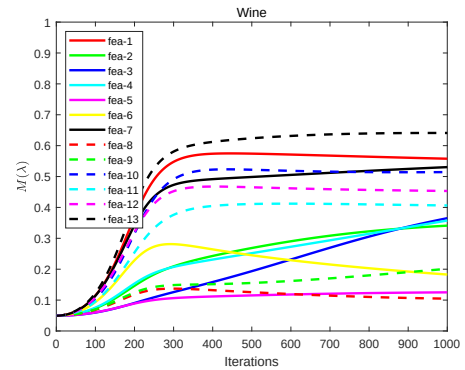
(a) $M(\lambda) = e^{-\lambda^2}$ with the learning rate 0.01.(b) $M(\lambda) = e^{-\lambda^2}$ with the learning rate 0.05.(c) $M(\lambda) = 1 - e^{-\lambda^2}$ with the learning rate 0.01.(d) $M(\lambda) = 1 - e^{-\lambda^2}$ with the learning rate 0.05.(e) $M(\lambda) = \frac{1}{1+e^{-\lambda}}$ with the learning rate 0.01.(f) $M(\lambda) = \frac{1}{1+e^{-\lambda}}$ with the learning rate 0.05.(g) Our proposed gate function, $M(\lambda) = \lambda\sqrt{e^{1-\lambda^2}}$, with the learning rate 0.01.(h) Our proposed gate function, $M(\lambda) = \lambda\sqrt{e^{1-\lambda^2}}$, with the learning rate 0.02.

Fig. S-1. The gate openings comparison of different gate functions on Wine dataset.

number of features selected is relatively high which varies from 45 to 63, while the number of data points per class is very low (varies from 8 to 13), which may not be adequate to represent the statistical characteristics of the classes very well. Such a situation may be avoided by using a validation data to make early stopping of the training process. But, this may not be a feasible option with such small training data. Another possible avenue, which we plan to explore, is the use of a suitable regularizer to avoid such a situation.

As an illustration of how the training process converges with iterations, we have randomly selected two low-dimensional and two-high dimensional datasets: Iris, Wine, SRBCT, and Leukemia. In Fig. S-2, we depict the variation of the mean square error with iterations for each of the three phases of training for the Iris, Wine, SRBCT, and Leukemia datasets, respectively.

It can be seen from these figures (Figs. S-2(a) - S-2(l)) that the training procedures in the feature selection and fine tuning phases converge faster than that of the rule extraction phase. In the rule extraction phase, the fuzzy rule base is extended to En-FRB, in which the number of rules becomes $(2D' + 1)S$. This reveals that in the rule extraction phase, there are more rules to be learned than that for each of the other two phases. Consequently, the rule extraction phase demands more training epochs than those by the other phases of the training.

TABLE S-II
THE TRAINING AND TEST ACCURACIES OF FSRE-AdATSK ON LOW-DIMENSIONAL DATASETS.

Datasets	Training accuracy	Test accuracy
Iris (4)	96.7	96.5
Appendicitis (7)	89.7	86.3
Pima (8)	76.5	75.7
Yeast (8)	59.5	58.1
Glass (9)	70.8	65.5
Page-blocks (10)	92.2	93.4
Wine (13)	99.3	97.3
Heart (13)	85.9	82.5
Wdbc (30)	96.8	95.4
Texture (40)	92.4	97.2
Spectfheart (44)	82.6	79.2
Sonar (60)	82.6	73.4

TABLE S-III
THE TRAINING AND TEST ACCURACIES OF FSRE-AdATSK ON HIGH-DIMENSIONAL DATASETS.

Datasets	FSRE-AdATSK (LSE)		FSRE-AdATSK (GD)	
	Training accuracy	Test accuracy	Training accuracy	Test accuracy
ORL (1024)	93.1	86.7	99.4	83.5
Colon (2000)	88.0	77.0	88.0	81.6
SRBCT (2308)	99.1	95.1	98.9	96.7
ARP (2400)	94.5	79.1	97.2	77.7
PIE (2420)	99.7	95.1	100.0	96.2
Leukemia (7129)	98.0	92.3	97.8	93.4
CNS (7129)	99.2	63.0	99.2	71.0

D. Adaptation of the Ada-softmin

We have explained in Section IV-B how the proposed AdaTSK classifier effectively solves the high-dimensional problems. Here, to illustrate that the proposed Ada-softmin can adaptively acquire $\hat{q}$ according to the current membership values, in Fig. S-3 we show the $\hat{q}$ used in different phases while dealing with the dataset, SRBCT. For high-dimensional datasets, as explained earlier, the centers are kept fixed in the first two phases: feature selection and rule extraction. The parameters of Ada-softmin are found to adapt to different $\hat{q}$ s for different instances before the training and maintained these values during the training. Note that, for every rule, there will be a set of computed values of $\hat{q}$ s, one for each instance. For a typical rule, the computed $\hat{q}$ s corresponding to all the training instances of SRBCT in the feature selection and rule extraction phases are shown in Fig. S-3(a) and Fig. S-3(b), respectively. On the other hand, in the fine tuning phase, the centers and the consequent parameters are trained together and consequently, the $\hat{q}$ s are changed along with iterations. The $\hat{q}$ s calculated for a typical rule for different instances just before the 100_{th} and the 200_{th} iterations are shown in Fig. S-3(c). The continuous line corresponds to the 100_{th} iteration and the dotted line corresponds to the 200_{th} iteration. This figure depicts how the $\hat{q}$ changes with iterations for different instances. Moreover, for a given rule and for one typical instance, the changes in $\hat{q}$ along with iterations are plotted in Fig. S-3(d).

E. The Robustness of AdaTSK

Here we investigate the robustness of a trained AdaTSK model with respect to perturbations of the centroids of the membership functions. For each of the 19 datasets, using a trained AdaTSK system we do two experiments. In the first

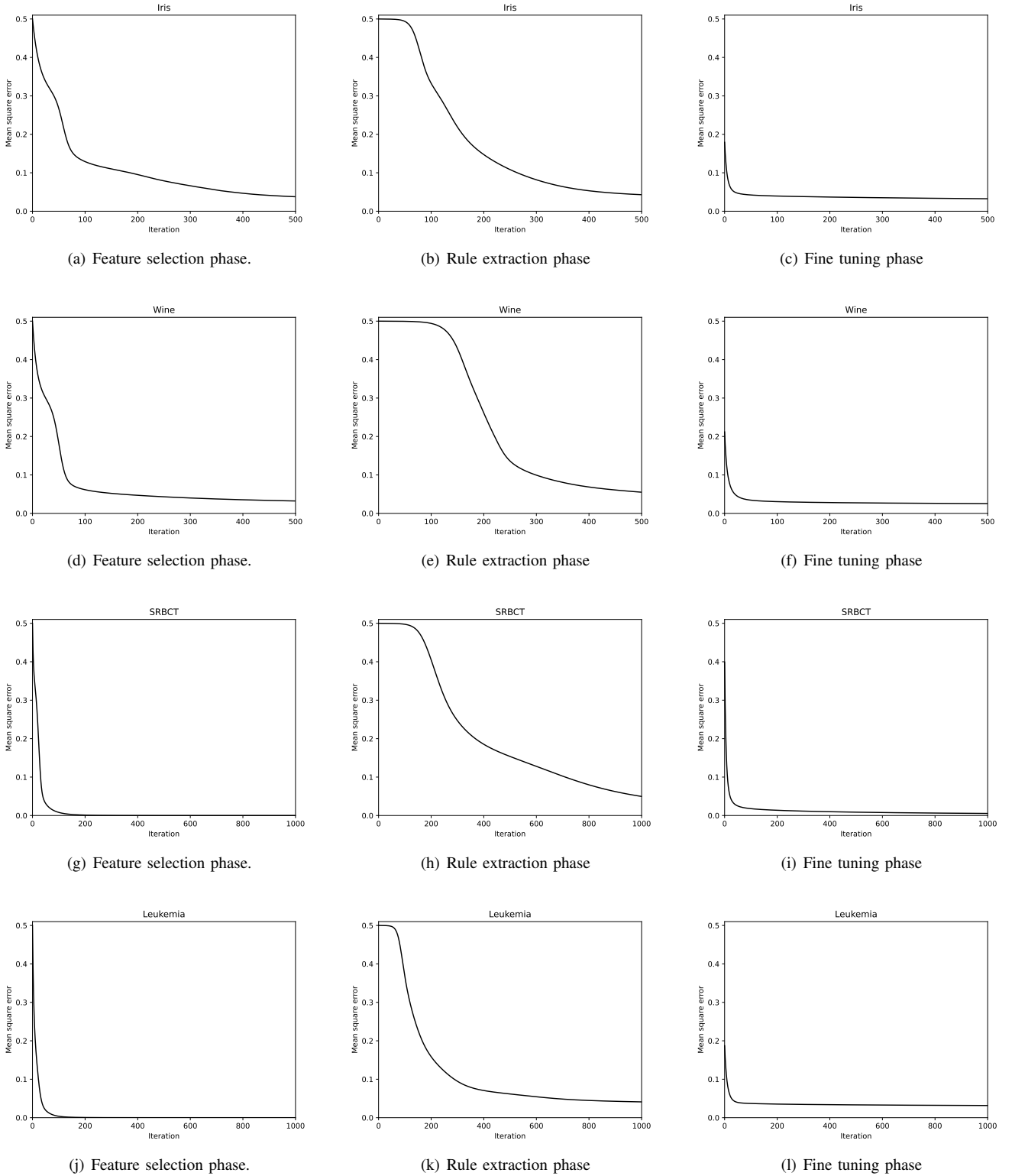


Fig. S-2. Mean square errors in the training phase.

experiment, we use additive Gaussian noise with mean zero and standard deviation 0.05 to the center of every antecedent membership function. This experiment is repeated five times. We report the average accuracies and standard deviations in columns two and three of Table S-IV. In the second experiment, we follow the same computational protocol, but we use additive Gaussian noise with a larger standard deviation, 0.10. The results of this experiment are summarized in columns four and five of the same table, Table S-IV. From Table S-IV, we observe that for several datasets there is a minor degradation in performance as expected. However, for some datasets, the average accuracy is found to be better than the performance of the

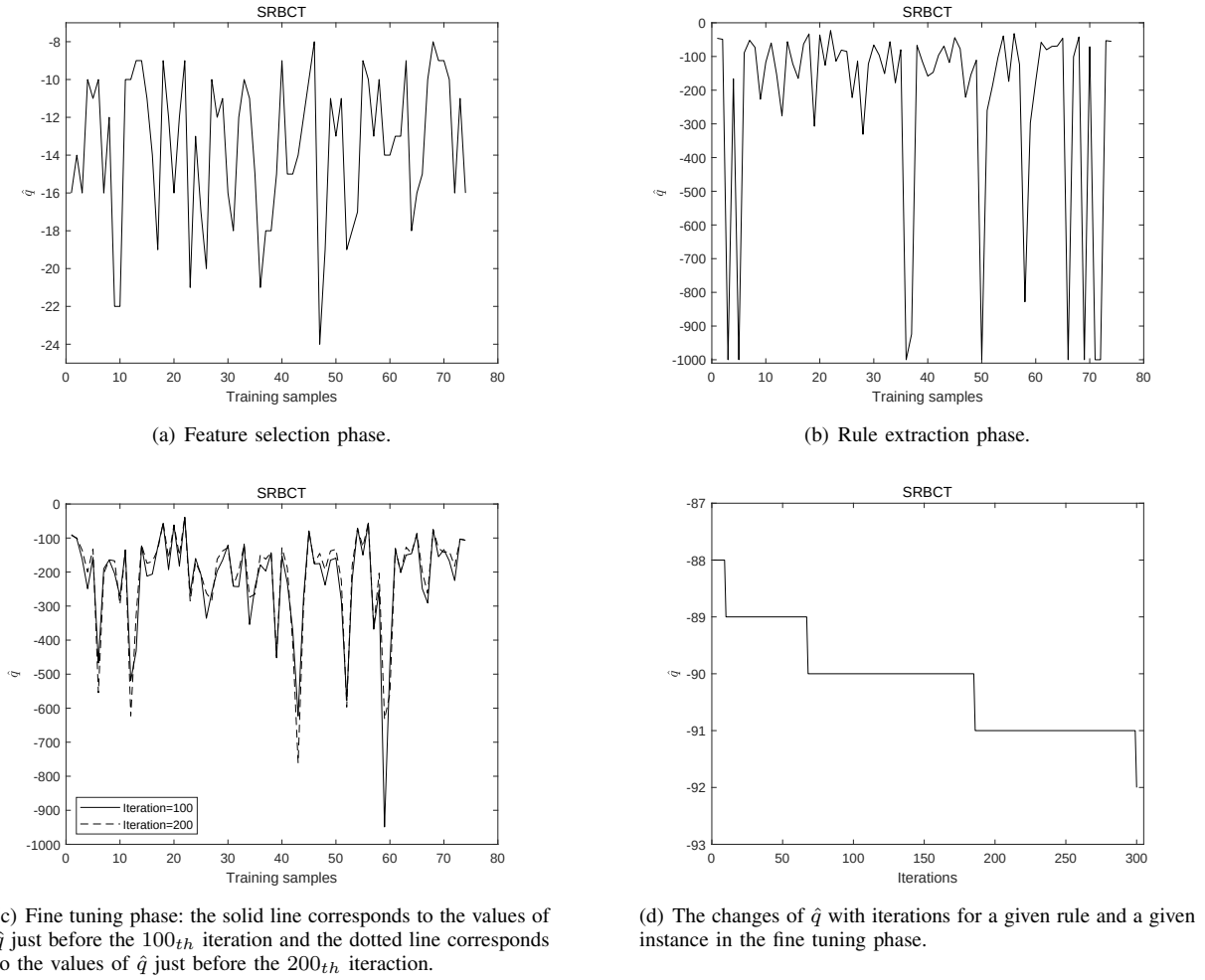


Fig. S-3. The $\hat{q}$ of Ada-softmin used on SRBCT dataset.

unperturbed AdaTSK (Table III). This is possible because the results reported in Table III and those in Table S-IV are based on different runs of the AdaTSK algorithm. In summary, AdaTSK is quite robust with respect to perturbations of the centroids of the membership functions.

F. Empirical comparison of computation time

Table S-V reports the computation time required by FSRE-AdaTSK and a standard neuro-fuzzy system both for the low-dimensional and high-dimensional datasets. Table S-V reveals that for low-dimensional datasets, the running time of feature selection is a little bit more than that of the standard neuro-fuzzy system. The total running time of FSRE-AdaTSK is longer than that of the standard neuro-fuzzy system. On the Texture dataset, the total running time of FSRE-AdaTSK is significantly higher than that of the standard neuro-fuzzy system and the main contributor to this is the rule extraction phase. There are three factors responsible for this. The complexity of the rule extraction phase is $O(D'^2CSN)$ where D' is the number of selected features. Although, the Texture dataset is in a lower dimension than that of the Sonar dataset, for example, the average number of features used in the rule extraction phase is 23.7 and that for Sonar is only 10.5. Moreover, the average number of rules for the Texture dataset is about five times of that for the Sonar dataset. Additionally, the Texture dataset has the highest number of instances. All these three components contribute to the high computational overhead for the Texture dataset. For high-dimensional datasets, the total running time is not much more than that of the standard neuro-fuzzy systems on some datasets, such as Colon, Leukemia and CNS. This happens because of a drastic reduction in the number of features in the rule extraction phase. Besides, for all of these datasets, the training time of the fine-tuning phase is less than that of the standard neuro-fuzzy systems, which demonstrates the effectiveness of the feature selection and rule extraction phases.

TABLE S-IV

ASSESSING ROBUSTNESS OF A TRAINED ADATSK SYSTEM WITH RESPECT TO PERTURBATIONS OF THE CENTROIDS OF THE MEMBERSHIP FUNCTIONS USING ADDITIVE GAUSSIAN NOISE WITH DIFFERENT STANDARD DEVIATIONS.

Dataset	Additive Gaussian noise with zero mean and standard deviation 0.05		Additive Gaussian noise with zero mean and standard deviation 0.10	
	Average accuracies	Standard deviations	Average accuracies	Standard deviations
Iris (4)	0.9533	0.0476	0.9267	0.0611
Appendicitis (7)	0.8682	0.0378	0.8636	0.0407
Pima (8)	0.7500	0.0352	0.7442	0.0359
Yeast (8)	0.5582	0.0325	0.5593	0.0319
Glass (9)	0.5442	0.0651	0.5512	0.0706
PageB (10)	0.9002	0.0057	0.9004	0.0059
Wine (13)	0.9694	0.0194	0.9722	0.0215
Heart (13)	0.8333	0.0483	0.8296	0.0509
Wdbc (30)	0.9553	0.0127	0.9509	0.0181
Texture (40)	0.9610	0.0065	0.9561	0.0081
Spectfheart (44)	0.7981	0.0660	0.7389	0.0874
Sonar (60)	0.7619	0.0337	0.7286	0.0490
ORL (1024)	0.9313	0.0384	0.9313	0.0384
Colon (2000)	0.6923	0.0973	0.6846	0.1058
SRBCT (2308)	0.8824	0.0696	0.8824	0.0789
ARP (2400)	0.9654	0.0363	0.9654	0.0363
PIE (2420)	0.9786	0.0249	0.9786	0.0249
Leukemia (7129)	0.8133	0.1024	0.8133	0.1024
CNS (7129)	0.6667	0.1988	0.6667	0.1988

TABLE S-V

COMPARISON OF TRAINING TIME OF STANDARD NEURO-FUZZY SYSTEM AND FSRE-ADATSK

Dataset	Standard	FSRE-AdaTSK			
		FS	RE	FT	Total
Iris (4)	1.05	1.29	1.28	0.91	3.49
Appendicitis (7)	1.03	1.23	1.84	0.69	3.75
Pima (8)	3.35	4.26	4.29	1.80	10.35
Yeast (8)	7.61	8.53	42.50	6.65	57.67
Glass (9)	2.59	2.60	4.05	1.36	8.02
Page-blocks (10)	28.63	30.17	87.41	16.90	134.48
Wine (13)	2.48	2.82	6.65	1.95	11.43
Heart (13)	3.39	3.67	8.09	1.32	13.09
Wdbc (30)	8.98	12.32	12.06	2.83	27.20
Texture (40)	111.66	123.96	917.53	51.79	1093.28
Spectfheart (44)	4.89	5.60	6.73	1.82	14.15
Sonar (60)	5.16	6.03	33.91	1.81	41.74
ORL (1024)	160.12	177.12	707.52	78.20	962.85
Colon (2000)	43.42	46.49	2.10	1.90	50.50
SRBCT (2308)	63.64	66.72	18.58	5.42	90.72
ARP (2400)	110.60	115.53	31.41	11.67	158.61
PIE (2420)	173.87	179.50	95.65	7.32	282.48
Leukemia (7129)	168.60	171.69	4.13	2.51	178.33
CNS (7129)	105.58	114.22	4.58	1.87	120.68